

PowerFix: Physics-Guided Iterative Repair for LLM-Assisted Power Converter Design

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Abstract

Large language model (LLM) agents are increasingly being explored for automated engineering design. However, repairing power-converter designs that fail engineering validation remains challenging: the system must identify physical causes, locate responsible design variables, and resolve violations without compromising other constraints. We introduce PowerFix, a physics-guided iterative repair framework that converts validation failures into structured diagnoses and executable patches while preserving topology and design requirements. PowerDiagGraph links failed checks to possible physical causes and constrains repair routes. An LLM prioritizes these routes, while a deterministic compiler translates selected repairs into authorized field-level edits. Candidates are validated on isolated design copies and committed only after all required engineering checks pass; failed validation can guide replanning. On our constructed 96-task flyback repair benchmark, comprising injected single faults, coupled faults, and held-out natural failures from our design generator, PowerFix achieves 78.7% verified repair success rate (VRSR), compared with 30.1% for the strongest of three LLM baselines. Three additional replanning rounds increase VRSR to 88.4%. These offline results provide pilot-stage evidence within a simulation-based design workflow, with hardware prototype validation and transfer to a second converter topology as the next steps toward deployment.

Introduction

Power converters connect energy sources and storage devices to electrical grids, regulating voltage, current, and power flow in renewable energy systems, electric transportation, and industrial automation (Bose 2009). Their failures are a major reliability concern. In 2023, Toyota recalled 43,442 plug-in hybrid electric vehicles in the United States due to a DC–DC converter defect that could cause overheating (the recall report was revised in 2024) (National Highway Traffic Safety Administration 2024). Beyond such safety risks, uncorrected engineering errors may cause severe hardware damage once a design is implemented. For example, an incorrect ripple-current calculation may result in the selection of a capacitor with an insufficient ripple-current rating. The converter may

continue to operate normally at first, while excessive ripple current increases the capacitor temperature and accelerates its degradation (Wang and Blaabjerg 2014).

Recent advances in large language models (LLMs) have driven growing interest in LLM-assisted power electronics design. PE-GPT explored LLM-based converter design (Lin et al. 2024, 2025), while Huang et al. proposed a multi-agent framework for controller tuning and calibration (Huang et al. 2025). LLM-based workflows nevertheless remain error-prone on engineering design tasks. On 269 switched-mode power-supply tasks, standalone GPT-4o and o3 failed on 85.6% and 74.6% of the cases, respectively; retrieval and iterative SPICE feedback increased o3’s solve rate to 90.7%, demonstrating the value of engineering evidence (Nau, Krummenauer, and Zimmermann 2026).

However, feedback-driven refinement does not by itself provide a structured repair process for a design that has already failed engineering validation. A repair system must connect observed violations to their underlying physical causes, identify the responsible design variables, restrict modifications to admissible fields, and verify that correcting the reported failure does not leave or introduce violations elsewhere. Existing workflows do not explicitly combine these requirements into a validation-gated repair procedure.

To address this problem, we introduce **PowerFix**, a physics-guided iterative repair framework for LLM-assisted power converter design. Its four-stage workflow consists of Failure Characterization, Graph-Guided Diagnosis, Constrained Repair Generation, and Isolated Validation.

The framework follows three principles.

1. Physics-grounded diagnosis. PowerFix converts validation failures into structured evidence and traces them through a topology-specific diagnostic graph to supported physical causes and admissible repair routes.

2. Constrained executable repair. An LLM prioritizes repair routes authorized by the diagnosis, while deterministic repair operators compile the selected routes into field-level edits. The LLM therefore does not directly perform unrestricted modification of the converter state.

3. Validation-before-commit. Each candidate patch is evaluated on an isolated copy of the current design and committed only if all required static, analytical, and circuit-simulation checks pass. Otherwise, the baseline remains unchanged and the validation evidence can guide a subsequent repair at-

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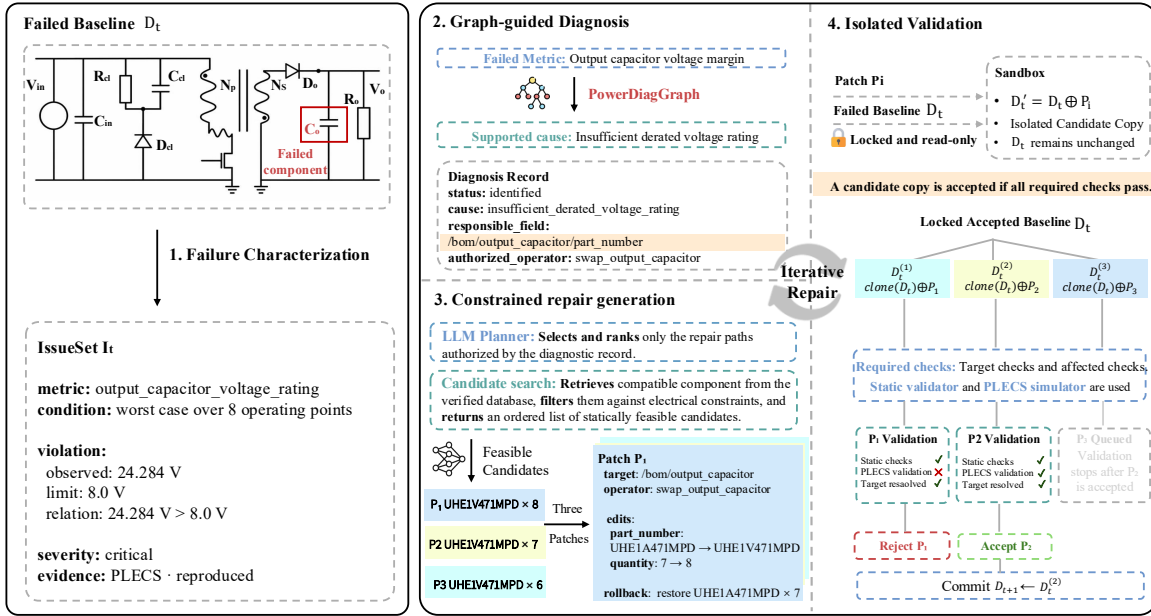


Figure 1: Overview of **PowerFix**. PowerFix performs iterative repair through failure characterization, graph-guided diagnosis, constrained repair generation, and isolated validation.

tempt.

Our contributions are threefold:

- We study how to repair failed converter designs so that they pass every required engineering check, and show why fixing the reported fault alone is insufficient.
- We introduce PowerFix, which links physics-guided diagnosis to authorized repair routes, combines LLM planning with deterministic patch compilation, and accepts patches only after they pass validation on isolated design copies.
- We build a 96-task flyback benchmark of single, coupled, and held-out natural failures, on which PowerFix reaches 78.7% group-weighted VRSR against 30.1% for the strongest of three LLM baselines (88.4% with three replanning rounds). Ablations test each stage; planning studies compare success and cost.

Related Work

Circuit and converter design. Converter optimization searches for trade-offs in efficiency, power density, and cost (Burkart and Kolar 2017); repair starts from an existing design that fails required checks. LLM-based assistance spans EDA scripts and circuit generation (Liu et al. 2023; Lai et al. 2025); Pan et al. (2025) survey the area. Power-electronics assistants support parameter and modulation design (Liu et al. 2024; Lin et al. 2024, 2025). Others gather requirements or tune controllers (Liu et al. 2026a; Huang et al. 2025). Simulation-guided refinement (Dong et al. 2026; Liu et al. 2026b) includes direct netlist edits (Nau, Krummenauer, and Zimmermann 2026); in PowerFix, the LLM selects routes allowed by the diagnosis and a compiler writes the edits.

Fault diagnosis. Gao, Cecati, and Ding (2015) review real-time fault diagnosis using models and signals. PowerDiag-Graph addresses failed design checks before hardware is built: it links analytical and simulation evidence to physical causes, which determine the permitted repair routes.

LLM feedback and program repair. Software patches can pass repair tests yet fail independent tests (Smith et al. 2015). RIRR captures a related gap: target fixes that still fail other engineering checks. Our Self-Refine baseline adapts iterative revision (Madaan et al. 2023) to engineering feedback; the Tool Agent uses actions and feedback, as in ReAct (Yao et al. 2023). Reflexion uses reflections to guide retries (Shinn et al. 2023); PowerFix uses validation feedback to re-plan within the existing diagnosis. SWE-bench, SWE-agent, and RepairAgent study repository repair (Jimenez et al. 2024; Yang et al. 2024; Bouzenia, Devanbu, and Pradel 2025). A software repair loop alone does not specify which physical changes are permitted or which checks those changes require.

The Repair Problem

Repair starts from a design that fails the full engineering regression. At step t , the failure set $\mathcal{I}_t$ (IssueSet in Figure 1) records each failed metric, its operating condition, observed and required values, and supporting evidence. Diagnosis and validation treat the failures as one set, because a fix for one can create another.

A patch P modifies only authorized design fields. Its required checks cover the reported failures and every check the modified fields can influence:

$$\mathcal{C}_{\text{req}}(P, \mathcal{I}_t) = \mathcal{C}_{\text{target}}(\mathcal{I}_t) \cup \mathcal{C}_{\text{affected}}(P). \quad (1)$$

The second set catches other failures that the patch may leave unresolved or introduce. Only a patch passing these checks can replace the current design. Appendix C gives the design state, patch representation, and full acceptance rule.

PowerFix

Figure 1 illustrates the four-stage repair loop.

Failure characterization. Patches may edit only primary fields, such as switching frequency, transformer winding parameters, and component selections. All other fields, including derived quantities, validation records, requirements, and the topology definition, are protected. Engineering tools recompute derived quantities. Failure characterization re-checks each reported violation against the current state and records it in the failure set $\mathcal{I}_t$.

Graph-guided diagnosis. PowerDiagGraph constrains the search for physical causes. Its nodes represent failed metrics, intermediate physical quantities, and possible causes. Arcs come from converter equations and design rules: they show how a metric decomposes into quantities or how a cause produces an effect. The flyback PowerDiagGraph contains 174 nodes and 301 arcs. It is authored offline once per topology and reused across tasks; its construction cost is excluded from the reported per-task costs. PowerDiagGraph is fixed during inference and cannot be extended by the model.

For each failed metric, diagnosis follows the corresponding subgraph to possible causes and the quantities needed to evaluate them. It retrieves design values and waveform features, then uses the topology’s analytical and simulation tools to compare measurements with physical predictions. Each cause is marked *supported*, *contradicted*, or *unresolved*. When the evidence cannot distinguish competing causes, PowerFix runs a targeted diagnostic simulation. The resulting record contains the causes, evidence, responsible primary fields, and allowed repair operations for this topology. Appendix D (“A worked diagnosis”) gives a MOSFET voltage-margin example in which simulation distinguishes insufficient clamp absorption from excessive leakage energy.

Constrained repair generation. A repair dimension is an aspect of the design, such as MOSFET selection. An operation specifies a permitted edit, such as swapping that MOSFET. An option is a concrete choice within a dimension, such as a catalog MOSFET. A repair route is one way of fixing the failure that the diagnosis supports; candidate patches instantiate it with particular parts or parameters.

An atomic patch records one permitted operation, its arguments, and the resulting field edits. Each edit names the target field and its current and proposed values. The planning LLM chooses which allowed repair routes to try first. A deterministic compiler produces the field edits and retrieves device properties from the component database when a part is replaced. Several atomic patches can be combined into a joint candidate for validation.

A hand-authored field-to-check map, separate from PowerDiagGraph, defines authorized fields and their associated checks. The validator takes the union of checks associated

with the fields written by a patch and rejects any patch that writes an unlisted field.

In our experiments, all 29 authorized fields map to the full engineering regression over eight operating points, so every candidate uses the same $\mathcal{C}_{\text{req}}$. Equation 1 permits patch-specific check selection, but we do not evaluate its potential savings. Appendix D describes an unused finer map with a different field schema.

Isolated validation. Each candidate is applied to a copy of the current design, leaving the baseline unchanged during validation. Checks run in order of increasing cost: static checks on schema, field permissions, units, preconditions, and component constraints; analytical checks on rating, duty-cycle, magnetic, and thermal constraints; and circuit simulation over the required operating conditions. The validation record includes the status of the target failures, check outcomes, newly introduced violations, and measured values. A candidate is committed only after every required check passes; otherwise it is rejected and the baseline remains unchanged.

The system retains rejected candidates and their validation evidence. When replanning is enabled, the planner reads this evidence and can choose another repair route within the existing diagnosis. Otherwise, the search continues through the preplanned candidates. The loop ends when a candidate is accepted or the repair budget is exhausted.

Evaluation Setup

We evaluate PowerFix on 96 failed flyback designs. The single-fault tasks cover every combination of three input specifications (85–132, 180–265, and 85–265 Vac), four output specifications spanning 10–24 W, and six fault classes ($3 \times 4 \times 6 = 72$). The fault classes are turns ratio, magnetizing inductance, magnetic realization, MOSFET, secondary rectifier, and output capacitor. Each task has a known violation and a recorded ground-truth repair. Twelve coupled tasks introduce interacting fault pairs, where the first visible violation does not identify the design error. The remaining twelve are *natural failures* from a design generator modeled on PEGPT (Lin et al. 2024). Its LLM roles propose the system, component, and magnetics designs; a fourth role integrates them, and engineering tools validate the result. We retain the generated design values, source outputs, and validation records. These tasks contain no injected faults and have no unique correct repair. We did not use them when developing or tuning PowerFix; they are scored on VRSR like the other two groups. Appendix A gives the construction.

We compare complete regeneration, revision from validation feedback, and revision through direct tool use (Appendix B). *Open-loop Regeneration* produces a complete replacement design. Our *Natural-language Self-Refine* baseline adapts iterative refinement (Madaan et al. 2023) to feedback from engineering checks; an LLM parser turns its free-text edits into field updates. The *Unstructured Tool-Feedback Agent* also calls the same component queries, analytical checks, and simulator as PowerFix, and edits fields based on what those tools return. None of the three uses PowerDiagGraph or constrained patches. All methods use

the same tasks, GPT-5 Mini-family backbone, component database, and validation conditions. Table 1 separates the primary comparison from the component and backbone studies. Each of the latter compares variants within its own protocol and full-system reference. Here R counts replanning rounds after the initial plan. After each method finishes, a deterministic *external scorer* checks its final design against the full engineering regression, using task records kept separately for evaluation. It is not an LLM judge, and its results are not fed into the repair loop.

The methods have different execution limits and do not use matched compute budgets. PowerFix makes about four times as many workflow simulation calls as the Tool Agent (Table 2). The primary comparison therefore measures complete workflows under their stated limits. Component effects are tested separately with a common candidate budget in the ablation.

Within each group, VRSR is the share of tasks whose final design resolves the target failure and passes the full engineering regression. The *Target Issue Resolution Rate* (TIRR) measures target resolution regardless of other checks. Both rates and per-task costs weight the three group means equally; we call these *group-weighted*. We define the repair-induced regression rate as $RIRR = 1 - \text{VRSR}/\text{TIRR}$. Error bars show 95% bootstrap intervals; differences use paired resampling. In the primary comparison, we resample base designs for controlled tasks and individual tasks for natural failures.

Results

Repair effectiveness. PowerFix reaches 78.7% VRSR against 30.1% for the strongest baseline (Figure 2), a gain of 48.6 percentage points (95% CI [33.8, 63.0]). It verifies 85 of 96 tasks (88.5% raw); group weighting gives equal importance to the two smaller groups. Thirty-one tasks are repaired by PowerFix alone (Figure 3(d)).

Self-Refine resolves more targets than the Tool Agent, but more of its target fixes still fail other checks; their ranking reverses under VRSR (Figure 2(b–d)). The largest gap is on coupled faults: 11 of 12 repairs versus two (Figure 3(b)). On natural failures, the paired interval for the gain is wide and includes zero ([−8.3, 58.3] percentage points), so we do not claim a significant gain.

Ablation. All four ablations share one full-system reference: 77 verified tasks and 72.7% VRSR (Figure 4). Under this protocol, the planner selects one or two dimensions from

Study	R	Candidates /task	Options /dimension	VRSR
Primary	0	≤ 6	≤ 48	78.7%
Ablation	0	≤ 6	≤ 8	72.7%
Backbone	5	≤ 6	≤ 8	75.9%

Table 1: PowerFix protocols on 96 tasks; VRSR uses GPT-5 Mini. Primary options apply to joint repairs (single-fault repair: up to three healthy candidates per route). Backbone: one candidate zero per round.

the smaller option pool in Table 1; the primary planner ranks all supplied routes. The option limit, selection policy, and order of candidate filtering differ between these studies, so their reference gap does not measure the effect of the option limit alone (Appendix E).

Removing graph-guided diagnosis causes the largest drop (53.2 points), with only one coupled repair and no natural repairs. Removing failure characterization or constrained repair generation costs 38–39 points. All three paired intervals exclude zero.

Removing isolated validation changes VRSR little but eliminates rejection: the system hands back a design for all 96 tasks, and the external scorer marks 22 invalid. The full system instead withholds 19 unrepaired outputs. Without this stage, an engineer would receive the invalid designs as completed repairs.

Repair-dimension selection. We compare the LLM planner with fixed-priority and random selection on 31 tasks that permit repairs in multiple dimensions. Each policy selects two permitted dimensions per round. Five seeds give 155 trajectories per policy (Appendix G).

The LLM planner has the highest VRSR at every reported budget and requests about a third fewer simulations at $R = 3$ (Figure 5). Recorded active repair time is longer than for the fixed policy (163 versus 125 seconds); it depends on cache reuse and execution order.

Repair budget. On the full 96-task benchmark, we vary the preplanned candidate-attempt limit A and the additional replanning limit R (Figure 6). These budgets separate searching within an initial plan from choosing another route after validation fails.

Three replanning rounds raise VRSR to 88.4%, verifying 91 of 96 tasks (Figure 6(a)). Most of the gain comes from three additional natural repairs: this 12-task group receives the same weight as the 72 single-fault tasks (Figure 6(b)). Two further rounds add simulation work without another verified repair. Appendix F lists counts and costs.

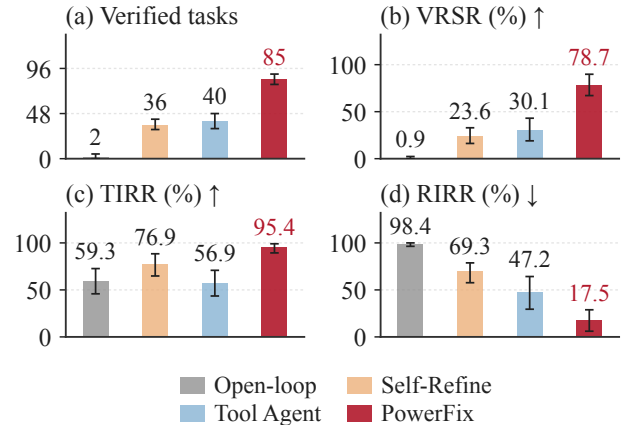
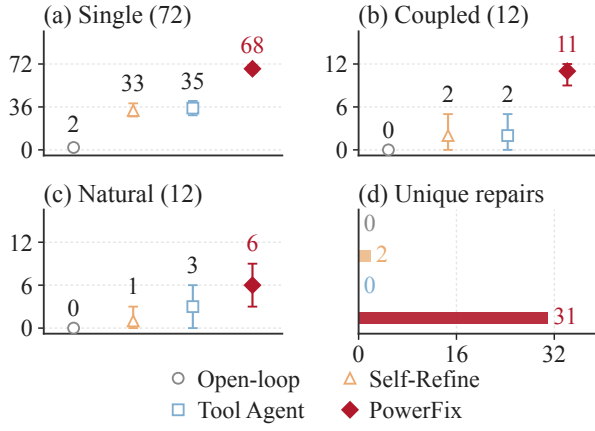


Figure 2: Primary results on 96 tasks: raw verified counts and group-weighted VRSR and TIRR. RIRR is the share of target fixes that fail other checks.



Backbone robustness. The backbone study follows the protocol in Table 1: $R = 5$, one candidate per round, and up to eight options per dimension. GPT-5 Mini provides its 75.9% reference.

Across twelve LLMs, VRSR spans 1.9 points (Figure 7; Appendix H); the lower end of each paired 95% interval stays

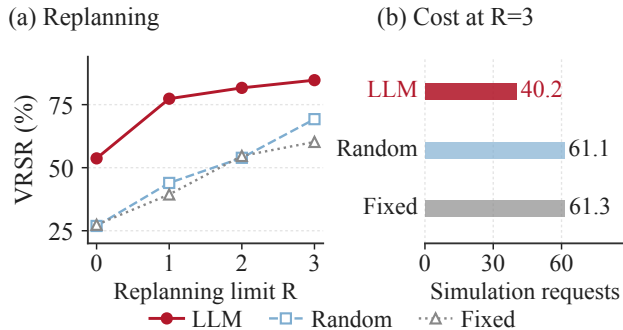
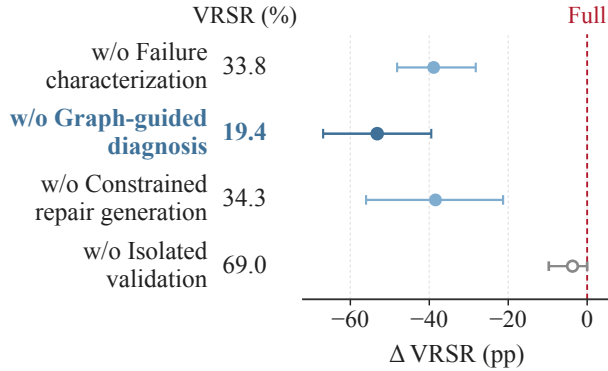


Figure 5: Repair selection (LLM: GPT-5 Mini): group-weighted (a) VRSR and (b) operating-point requests at $R = 3$, including cache hits.

within five points of the reference. Scorer outcomes agree on 91 of 96 tasks: only five single-fault tasks differ. All models leave the same two coupled and five natural tasks unverified and request about 18 to 19 operating-point evaluations per task.

Cost. PowerFix shifts work toward simulation: about 3% of the Tool Agent’s tokens but four times its workflow simulations (Table 2). At $R = 0$, one planning call orders the routes; enumeration, patch compilation, and validation then run deterministically. Mean LLM calls rise to 1.54 at $R = 3$ and 1.77 at $R = 5$. Costs include failed tasks but exclude initial graph and authorization-map authoring. Deployment needs a setup budget for each topology and ongoing capacity for updates and simulation.

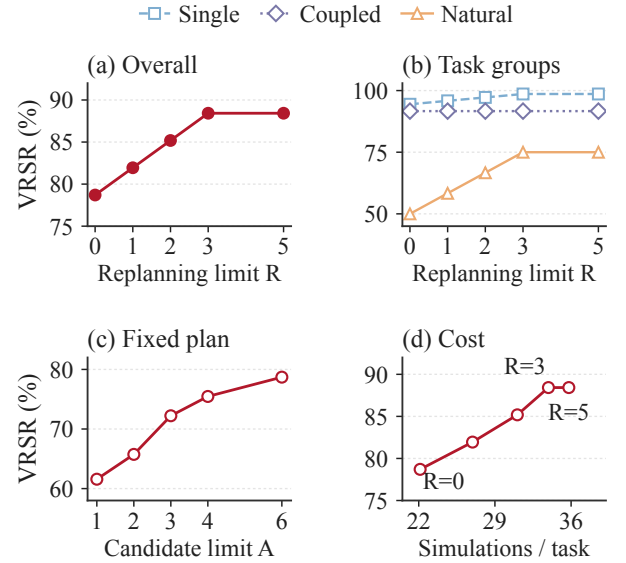


Figure 6: Budget effects on 96 tasks: (a) overall and (b) group repair rates with replanning; (c) candidate attempts at $R = 0$; (d) replanning versus mean workflow simulations. Axis ranges differ.

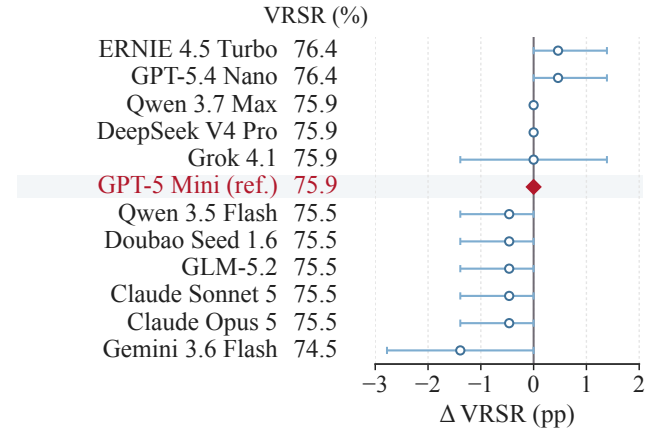


Figure 7: Backbone differences from its own GPT-5 Mini reference (75.9% VRSR; $R = 5$; Table 1).

Table 2 counts logged simulation calls; Figure 5 reports requested operating-point evaluations, including cache hits (Appendix E). These counts are not interchangeable.

Method	Simulations	LLM calls	Tokens (k)
Open-loop	1.74	4.00	37.10
Self-Refine	2.84	5.23	33.44
Tool Agent	5.31	11.08	82.86
PowerFix	22.13	1.00	2.73

Table 2: Group-weighted primary cost per task ($R = 0$ for PowerFix), including failures. Simulation and LLM counts are workflow requests (Appendix J).

Remaining failures. Five tasks remain unverified at $R = 3$: one single-fault, one coupled, and three natural. At $R = 0$, the one unverified coupled case combines magnetizing-inductance and MOSFET faults in a 12 V, 2 A design at 85–132 Vac. None of its three candidate bundles passes the full engineering regression. One resolves the reported targets but still violates the output-ripple limit at low line and full load. This case exposes a limit of the available repair routes. Natural failures are also difficult: they have no unique ground-truth repair and can admit several plausible causes. Self-Refine’s free-form edits verify three tasks missed by PowerFix at $R = 0$.

Simulation-Stage Pilot and Deployment Path

Use in a design workflow. PowerFix runs after an LLM-generated design fails the full engineering regression. It hands the engineer a verified patch or a rejection with supporting evidence; the engineer retains sign-off authority. Each patch records the fields written, their old and proposed values, the operation used, and the rollback edits that restore the old values. Requirements and topology definitions remain protected; the team approves the editable-field set before running repairs.

For the output-capacitor example in Figure 1, the accepted record (P_2) replaces UHE1A471MPD with UHE1V471MPD and explicitly retains quantity 7. This unchanged quantity counts as a write in the profile below. The patch detail box in Figure 1 shows the rejected P_1 , which uses eight capacitors and fails simulation on a copy; the baseline stays unchanged.

Patch size. Among 85 accepted submissions at $R = 0$, counting each joint repair once, 41 explicitly write at most two distinct fields, 41 write 25–26, and three write three. All large submissions include magnetics redesign, which writes 25 core, winding, and primary-clamp fields; seven also write a MOSFET or rectifier catalog reference. All writes are authorized, and every submission includes rollback edits. Appendix K details the distribution, operation coverage, and counting rules. Review time and engineer acceptance remain unmeasured.

Simulation-stage pilot. Engineers use simulation to refine converter designs and check operating limits before building hardware. PowerFix automates repair and revalidation within

this stage. We evaluated it offline on the same 12 held-out failures from design-generator runs: it repaired 6 (50%) under the primary protocol and 9 (75%) with three replanning rounds. Each repaired design passed the full engineering regression across eight operating points. These results measure recovery within the simulation workflow; engineer acceptance and hardware performance remain untested. We plan two next steps.

(i) *Hardware prototype validation.* We will build the repaired 24 V, 1 A flyback in Figure 1 and measure peak device voltage, output ripple, and efficiency across the same eight operating points. We will compare measurements with simulation predictions and engineering limits to test whether the repaired hardware performs as predicted. Discrepancies will guide refinements to the simulation models and checks.

(ii) *Transfer to another converter topology.* We will extend PowerFix to boost or LLC by building its PowerDiagGraph, registering repair operations, and checking the editable-field authorization map. We will measure repair success and simulation cost on failed designs and report adaptation effort in person-days, identifying which graph nodes and rules were reused or newly authored.

Deployment risk. A patch can pass every encoded check yet miss a requirement, such as supplier lead time. Authorization limits where patches can write, and rollback can undo them. Neither guarantees that the encoded checks are complete. Engineer review and hardware checks remain necessary before release.

Limitations

Results cover flyback designs and the encoded checks; the twelve natural failures come from one generator. We cannot fully verify model access across methods. Records confirm a shared endpoint and requested alias for 24 of 96 primary tasks; the other 72 are matched only at the documented model-family level (Appendix I). On those 24 tasks, PowerFix verifies 17 designs and the Tool Agent five. The separate backbone study spans 1.9 VRSR points across twelve models. Engineer review time and hardware performance remain unmeasured.

Conclusion

PowerFix treats repair as more than resolving the reported fault: every required engineering check must also pass. It combines physical diagnosis, authorized edits, and validation before acceptance, reaching 78.7% group-weighted VRSR against 30.1% for the strongest baseline.

The ablation distinguishes finding repairs from rejecting invalid outputs. Removing diagnosis costs 53.2 VRSR points; removing isolated validation costs only 3.7 but hands back 22 invalid designs out of 96. Success rates alone do not show which invalid outputs were withheld.

All accepted patches stay within 29 authorized fields and include rollback. Their records show engineers the proposed edits and supporting checks, while leaving sign-off to them.

The evidence is offline and limited to flyback. The deployment plan will test whether simulated repairs translate to hardware and other converter designs.

References

- Bose, B. K. 2009. Power Electronics and Motor Drives—Recent Progress and Perspective. *IEEE Transactions on Industrial Electronics*, 56(2): 581–588.
- Bouzenia, I.; Devanbu, P. T.; and Pradel, M. 2025. RepairAgent: An Autonomous, LLM-Based Agent for Program Repair. In *2025 IEEE/ACM 47th International Conference on Software Engineering (ICSE)*, 2188–2200.
- Burkart, R. M.; and Kolar, J. W. 2017. Comparative η - ρ - σ Pareto Optimization of Si and SiC Multilevel Dual-Active-Bridge Topologies With Wide Input Voltage Range. *IEEE Transactions on Power Electronics*, 32(7): 5258–5270.
- Dong, M.; Li, H.; Yin, S.; Lin, J.; Zhang, H.; Zeng, Z.; and Cheng, Y. 2026. A LLM-Guided Approach for Active-Clamped Flyback Converter of Power Processing Units in Electric Propulsion. *IET Electric Power Applications*, 20(1): e70154.
- Gao, Z.; Cecati, C.; and Ding, S. X. 2015. A Survey of Fault Diagnosis and Fault-Tolerant Techniques—Part I: Fault Diagnosis With Model-Based and Signal-Based Approaches. *IEEE Transactions on Industrial Electronics*, 62(6): 3757–3767.
- Huang, D.; Chen, Y.; Li, J.; and Dai, N. 2025. An LLM-Powered Multi-Agent Framework for Automated Controller Tuning and Calibration. In *2025 IEEE 3rd International Power Electronics and Application Symposium (PEAS)*, 1596–1601.
- Jimenez, C. E.; Yang, J.; Wettig, A.; Yao, S.; Pei, K.; Press, O.; and Narasimhan, K. 2024. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? In *The Twelfth International Conference on Learning Representations*.
- Lai, Y.; Lee, S.; Chen, G.; Poddar, S.; Hu, M.; Pan, D. Z.; and Luo, P. 2025. AnalogCoder: Analog Circuit Design via Training-Free Code Generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, 379–387.
- Lin, F.; Li, X.; Lei, W.; Rodriguez-Andina, J. J.; Guerrero, J. M.; Wen, C.; Zhang, X.; and Ma, H. 2025. PE-GPT: A New Paradigm for Power Electronics Design. *IEEE Transactions on Industrial Electronics*, 72(4): 3778–3791.
- Lin, F.; Liu, J.; Li, X.; Zhao, S.; Zhao, B.; Liao, X.; Ma, H.; and Zhang, X. 2024. PE-GPT: A Physics-Informed Interactive Large Language Model for Power Converter Modulation Design. In *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, 1744–1747.
- Liu, J.; Lin, F.; Li, X.; Zhao, S.; and Lim, K. H. 2026a. Physics-Informed Autonomous LLM Agents for Explainable Power Electronics Modulation Design. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(47): 40063–40072.
- Liu, M.; Ene, T.-D.; Kirby, R.; Cheng, C.; Pinckney, N.; Liang, R.; et al. 2023. ChipNeMo: Domain-Adapted LLMs for Chip Design. *arXiv preprint arXiv:2311.00176*.
- Liu, Q.; Feng, J.; Yuan, X.; Busquets-Monge, S.; Guo, X.; Zhang, Y.; Zhang, Y.; and Wang, K. 2026b. An Accessible Large-Language-Model-Based Framework for Power Converter Design. *IET Power Electronics*, 19(1): e70259.
- Liu, Q.; Feng, J.; Yuan, X.; Zhang, Y.; Busquets-Monge, S.; and Wang, K. 2024. Using ChatGPT for the Knowledge and Design of Power Electronics Converters. In *2024 IEEE 9th Southern Power Electronics Conference (SPEC)*, 1–6.
- Madaan, A.; Tandon, N.; Gupta, P.; Hallinan, S.; Gao, L.; Wiegrefe, S.; Alon, U.; Dziri, N.; Prabhume, S.; Yang, Y.; Gupta, S.; Majumder, B. P.; Hermann, K.; Welleck, S.; Yazdanbakhsh, A.; and Clark, P. 2023. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems*, volume 36, 46534–46594. Curran Associates, Inc.
- National Highway Traffic Safety Administration. 2024. Part 573 Safety Recall Report 23V-478. Technical Report 23V-478, U.S. Department of Transportation, Washington, DC. Submitted by Toyota Motor Engineering & Manufacturing.
- Nau, S.; Krummenauer, J.; and Zimmermann, A. 2026. Evaluating LLM-Based Workflows for Switched-Mode Power Supply Design. *ACM Transactions on Design Automation of Electronic Systems*. Just accepted.
- Pan, J.; Zhou, G.; Chang, C.-C.; Jacobson, I.; Hu, J.; and Chen, Y. 2025. A Survey of Research in Large Language Models for Electronic Design Automation. *ACM Transactions on Design Automation of Electronic Systems*, 30(3).
- Shinn, N.; Cassano, F.; Gopinath, A.; Narasimhan, K.; and Yao, S. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing Systems*, volume 36, 8634–8652. Curran Associates, Inc.
- Smith, E. K.; Barr, E. T.; Le Goues, C.; and Brun, Y. 2015. Is the Cure Worse Than the Disease? Overfitting in Automated Program Repair. In *Proceedings of the 2015 10th Joint Meeting on Foundations of Software Engineering*, 532–543. Association for Computing Machinery.
- Wang, H.; and Blaabjerg, F. 2014. Reliability of Capacitors for DC-Link Applications in Power Electronic Converters—An Overview. *IEEE Transactions on Industry Applications*, 50(5): 3569–3578.
- Yang, J.; Jimenez, C. E.; Wettig, A.; Lieret, K.; Yao, S.; Narasimhan, K.; and Press, O. 2024. SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering. In *Advances in Neural Information Processing Systems*, volume 37, 50528–50652. Curran Associates, Inc.
- Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K.; and Cao, Y. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *The Eleventh International Conference on Learning Representations*.

A Benchmark Construction

Table S1 defines the specification and fault codes used in the task records. Each healthy input–output combination is used once for each fault class. Within every class, four tasks are mild, four moderate, and four severe. The injected fields and reference repairs are kept in private annotations. Reported failures and their evidence remain available to the methods.

Table S1: Independent choices in the controlled benchmark. Any input can be paired with any output and fault class.

Code	Specification or fault
I1	85–132 Vac
I2	180–265 Vac
I3	85–265 Vac
O1	5 V / 2 A (10 W)
O2	12 V / 1 A (12 W)
O3	12 V / 2 A (24 W)
O4	24 V / 1 A (24 W)
F1	Transformer turns-ratio selection
F2	Magnetizing-inductance selection
F3	Magnetic-component realization
F4	MOSFET selection
F5	Secondary-rectifier selection
F6	Output-capacitor selection

Construction checks. The retained construction reports record 16 PLECS executions for the healthy bases and 1,512 for controlled-fault construction. The latter rejected 1,309 candidate faults before freezing the 72 tasks. Rejection reasons include a missing target violation, unrelated critical failures, and failed or incomplete validation; a candidate can have several reasons. The packaging audit checked all 72 public tasks and their evidence hashes and found no private ground-truth files in the public copies. These are construction costs, excluded from the repair costs in Table 2.

Coupled faults. Table S2 lists the four interaction patterns. Each pattern is tied to one output specification and instantiated at all three input ranges. Thus the data do not separate the effect of fault-pair type from that of output specification. Coupled and natural tasks are shuffled together before assigning T0073–T0096; identifiers do not encode the group.

Table S2: Coupled-fault patterns. Each row contributes three tasks, one at each input range I1–I3.

Output	Faults	Interaction
O1	F5+F6	Rectifier loss and output ripple
O2	F1+F4	Turns ratio and MOSFET voltage stress
O3	F2+F4	Peak current, MOSFET loss, and temperature
O4	F1+F5	Primary-device stress and rectifier reverse voltage

Natural-failure records. The source runs were completed on 22 July 2026 (UTC), with one retained failed output for each input–output specification. The source-to-task mapping

and the original design and validation records are retained. Preparing benchmark copies normalizes constraint labels and simulator bindings; it does not inject faults. The selection configuration requires an observed engineering failure and prevents access to private reference repairs. These tasks were not used to develop or tune PowerFix. The supplied records identify the selected designs, but do not enumerate the full source pool and every exclusion. This is a held-out set of log-derived failures, not a random sample of all generator failures.

B Baselines and Execution Limits

Table S3 compares the repair mechanisms and execution limits. These are not matched compute budgets.

Table S3: Primary repair mechanisms and execution limits. All methods undergo external scoring. PowerFix uses $R = 0$; validation tries preplanned candidates without another LLM planning call.

Method	Repair mechanism	Limit
Open-loop	Regenerates the full design.	Three proposal calls and one integration call.
Self-Refine	Revises from validation feedback; an LLM parses the edits.	At most three refinement rounds.
Tool Agent	Calls engineering tools and edits fields directly.	At most 16 agent steps.
PowerFix	Uses PowerDiagGraph to limit repairs; a compiler writes the edits.	One plan and at most six candidate validations.

For primary single-fault repair, PowerFix can use component and magnetics routes. Each route retains up to three candidates after deterministic health filtering. The planner ranks all supplied routes. Coupled and natural tasks instead use joint candidates, drawing from up to 48 options per dimension. Validation stops at the first accepted candidate, so six is an upper bound rather than the number executed on every task.

The ablation and backbone studies cap each dimension at eight options before candidate construction and health filtering. The ablation selects one or two permitted dimensions at $R = 0$; the backbone study validates one candidate per round at $R = 5$. Their full-system references therefore differ in route selection, filtering order, and available candidates, as well as the option cap. Appendix E examines the observed task differences. Appendix F gives the budget checkpoints.

Scoring and recorded randomness. The external scorer checks sealed waveform hashes and recomputes metrics instead of trusting saved pass/fail flags. It starts PLECS when usable records are absent. Static or analytical failures can stop scoring before simulation. Its outputs are kept outside the repair loop.

The primary comparison retains one trajectory per method–task pair. The base seed is 20260721; task-specific seeds are derived from benchmark position and accompany

the model requests. Task bootstrap intervals do not measure repeated decoding on the same task. The resampling units are specified in Appendix E; model-access records and prompt contents are described in Appendices I and J.

C Full Problem Formalization

We give the design state and acceptance rule behind Equation 1. Let D_t be the converter design state at step t , containing its topology, editable parameters, constraints, analytical results, and simulation waveforms. Applying patch P produces the candidate $D_P = D_t \oplus P$.

Let

$$\mathcal{I}_t = \{\iota_1, \iota_2, \dots, \iota_m\} \quad (2)$$

denote the structured failure representations associated with D_t ; each ι records a failed metric, its operating condition, the observed value, the required target, and supporting evidence.

For converter topology T , the topology-specific repair environment is

$$\mathcal{E}_T = (\mathcal{G}_T, \Omega_T, \mathcal{B}, \mathcal{V}_T), \quad (3)$$

where $\mathcal{G}_T$ is PowerDiagGraph, Ω_T the permitted repair operations, $\mathcal{B}$ the component database shared across topologies, and $\mathcal{V}_T$ the executable analytical and simulation tools. PowerDiagGraph is defined as

$$\mathcal{G}_T = (\mathcal{N}_T, \mathcal{A}_T), \quad (4)$$

with $\mathcal{N}_T$ containing failed metrics, intermediate physical quantities, and candidate causes, and $\mathcal{A}_T$ containing decomposition and causal relations such as *decomposed into*, *determined by*, and *caused by*.

An atomic patch is

$$P_t = (o_t, a_t, \Delta_t), \quad (5)$$

with $o_t \in \Omega_T$ the selected operation, a_t its arguments, and Δ_t the resulting field-level modifications. Validating the candidate produces

$$R_P = \text{Validate}(D_P, \mathcal{C}_{\text{req}}(P, \mathcal{I}_t)), \quad (6)$$

and acceptance is

$$\text{Accept}(P) = \text{Valid}(P) \wedge \text{ValidationPassed}(R_P), \quad (7)$$

where $\text{Valid}(P)$ requires the prescribed format and preconditions and edits confined to authorized fields, and $\text{ValidationPassed}(R_P)$ requires every check in $\mathcal{C}_{\text{req}}$ to have been executed and to satisfy its acceptance criterion. The state update is

$$D_{t+1} = \begin{cases} D_t \oplus P, & \text{if } \text{Accept}(P), \\ D_t, & \text{otherwise.} \end{cases} \quad (8)$$

D Method Details

PowerDiagGraph composition. The flyback PowerDiagGraph contains 174 nodes and 301 directed arcs: 32 failed-metric, 76 mechanism, 14 observed-evidence, 25 design-field, 10 diagnostic-test, 10 validation-check, and 7 repair-operation nodes. It is authored once per topology and fixed during inference. The seven operation nodes describe topology-level repairs; they are not an enumeration of the

four experimental atomic operators, the four selection dimensions in the backbone study, or the nine Top-2 dimensions. These representations have different roles and granularity. The four atomic operators replace a MOSFET, rectifier, or output-capacitor bank, or redesign the magnetics. Likewise, the 25 graph field nodes describe diagnostic relationships, not the 29 editable paths in the experimental authorization map. Graph check nodes support diagnosis; the runtime map determines which checks execute.

A worked diagnosis. Table S4 gives the numerical evidence used by the diagnostic test fixture. This is an illustrative calculation, not an additional benchmark task. The initial values support insufficient voltage margin and excessive turn-off overshoot, while the plateau residual contradicts excessive reflected voltage. Leakage and clamp absorption remain unresolved until the additional measurements are supplied. With $L_{lk} = 200 \mu\text{H}$ and turn-off current 10 A, leakage energy is 10 mJ. Overshoot also rises by 27.4 V per ampere between the two load measurements, above the rule’s 10 V/A threshold. Together these rules identify excessive leakage energy as the primary cause and also support insufficient clamp absorption. The graph requests a registered leakage/clamp discrimination experiment when these values are missing; the fixture supplies them to test the rule outcomes.

Table S4: Numerical MOSFET example. Thresholds belong to this fixture; they are not universal device limits.

Quantity	Value	Required
Rating minus peak	$650 - 742 = -92 \text{ V}$	$\geq 65 \text{ V}$
Plateau residual	$ 505 - 495 = 10 \text{ V}$	$\leq 25 \text{ V}$
Turn-off overshoot	$742 - 505 = 237 \text{ V}$	$\leq 120 \text{ V}$
Leakage energy	10 mJ	$\leq 4 \text{ mJ}$
Clamp absorption fraction	$4/10 = 0.40$	≥ 0.80

The field-to-check map. The experimental map has 58 associations over 29 paths. Each path maps to both target-issue resolution and the full engineering regression, so every candidate faces the same eight-operating-point check set. The patch-dependent definition in Equation 1 also permits finer selection. A separate map contains 43 checks and 368 associations over 26 paths, but uses a different field schema and coverage: its paths are not the experimental whitelist minus three entries. The schemas and coverage would have to be reconciled before substitution. This alternative is unused in the reported results, and no simulation saving from selective checks is measured. The reported localization concerns edits, not a reduced regression.

Interfaces and decision owners. Table S5 separates model decisions from the programs that produce and check concrete edits.

Compilation preconditions. Before simulation, the compiler and operator catalog check the base-state hash, operator registration and version, topology and design-family scope, exact argument names, and numeric bounds. They require the contract’s component facts and any physical-applicability evidence. Duplicate paths, missing or extra writes, and writes

Table S5: Module interfaces. All rows except planning are deterministic; PLECS supplies simulation evidence where needed.

Module	Input → output or decision
Characterization	Design and reported failures → values, limits, conditions, and evidence links in a failure set.
Diagnosis	Failure set and fixed rules → supported, contradicted, or unresolved causes and available repairs.
LLM planning	Diagnosis and legal choices → route ordering or dimension selection.
Synthesis	Selected choices and frozen catalogs → concrete candidate intents.
Compilation	Intent and current state → exact field edits and their reversals.
Validation/gate	Patch, map, and design copy → check results and commit decision.
External scorer	Final output and separate task records → evaluation, never repair feedback.

to protected or derived fields are rejected. Every target must resolve to an existing scalar field. Compilation records its old value, inverse edit, and specification/topology hashes; the gate checks those invariants.

Recorded trace: T0053. The 24-V, 1-A task links its reproduced capacitor-margin failure to a state hash and eight sealed waveform records. The public limit is $10 \times 0.8 = 8$ V; the diagnostic rule uses the observed maximum 24.28418 V and obtains a margin of -16.28418 V. It identifies the voltage-rating cause without an additional diagnostic experiment. The LLM selects the component route. Three candidates are compiled; two are validated before acceptance. The first fails the steady-state drift check at both low-line, minimum-load bus corners: 0.66770 and 0.66530 V, against 0.24 V. The second passes the full regression. Its analytical ripple estimate is 0.01646 V; the maximum simulated, linearly detrended ripple is 0.069509 V, also below 0.24 V. Its explicit two-field write set includes the unchanged quantity $7 \rightarrow 7$; only one field changes value. Rollback records both fields, distinguishing the write footprint from the changed-value count.

The formal run records 36 PLECS executions: 12 for failure reproduction and 12 for each validated candidate. Each sweep covers eight corners; four light-load corners require longer settling and a second execution. The final external scorer checks hashes and recomputes metrics from sealed waveforms, making zero fresh simulation calls in this case. These counts use the completed run and its validation reports, not three earlier rejected events retained in the same history.

E Detailed Evaluation and Accounting

Fault classes. Table S6 separates the single-fault results by the injected design error. Each column covers all twelve input/output specifications, so differences are not caused by unequal class sizes. The classes differ in repair difficulty: baseline coverage is especially low for turns ratio and mag-

netizing inductance. These counts describe this constructed set, not the frequency of faults in service.

Table S6: Verified single-fault tasks by class, out of twelve per column. Fault codes are defined in Table S1.

Method	F1	F2	F3	F4	F5	F6
Open-loop	0	0	1	1	0	0
Self-Refine	3	3	4	4	12	7
Tool Agent	0	1	9	4	10	11
PowerFix	10	11	12	11	12	12

Which methods repair the same tasks. Table S7 partitions all 96 tasks by the exact set of methods that verify them. The intersections show both shared coverage and repairs missed by PowerFix: Self-Refine alone verifies two single-fault tasks, and Self-Refine and the Tool Agent verify one natural task that PowerFix does not. The empty-set row contains tasks no method verifies.

Table S7: Exact repair intersections. Only the listed methods verify a task in each row; S, C, and N denote single, coupled, and natural tasks.

Methods that verify	S	C	N
PowerFix	19	8	4
Tool Agent, PowerFix	17	1	2
Self-Refine, Tool Agent, PowerFix	18	1	0
Self-Refine, PowerFix	12	1	0
None	2	1	5
Self-Refine	2	0	0
Open-loop, Self-Refine, PowerFix	1	0	0
Open-loop, PowerFix	1	0	0
Self-Refine, Tool Agent	0	0	1
Total	72	12	12

New violations versus remaining violations. RIRR counts resolved targets whose final designs still fail other checks; those failures need not be new. A stricter audit compares the initial and final failed-check sets. If these are F_{initial} and F_{final} , a newly failed non-target check belongs to

$$F_{\text{final}} \setminus (F_{\text{initial}} \cup C_{\text{target}}).$$

The initial set combines observed failures with initial qualification failures. For single-fault tasks, target exclusion also matches the expected-check family; coupled and natural tasks use normalized target identifiers. Check matching removes only terminal run suffixes such as $:r1$ and $:x1$. It retains the metric and its line, load, and bus condition, so a failure at a new operating corner counts. Table S8 uses raw task counts rather than group-weighted rates.

The three PowerFix cases are T0086, T0088, and T0096: rejected attempted designs retained for external scoring, not accepted patches. Four single-fault tasks (T0037, T0045, T0052, T0065) have no final design for this audit and are marked unavailable, not zero. Their default `frozen_field_violation` flags therefore do not

Table S8: Tasks with at least one newly failed non-target check. Denominators include only records with an evaluable final design.

Method	Single	Coupled	Natural	Total
Open-loop	47/72	9/12	8/12	64/96
Self-Refine	36/72	8/12	10/12	54/96
Tool Agent	21/72	6/12	5/12	32/96
PowerFix	0/68	1/12	2/12	3/92

demonstrate four actual writes to protected fields. This audit is separate from the validation-removal experiment in the main text.

Ablation reference and route differences. Table S9 gives the group counts behind the ablation. The seven single-fault tasks repaired by the primary system but not its ablation reference are all turns-ratio faults. On T0003, T0028, T0048, T0055, T0064, and T0071, the primary system uses magnetics repair; the ablation reference selects only rectifier replacement. An offline reconstruction with the current generator finds no healthy rectifier-only candidate even with all eleven rectifier options.

For T0033, the ablation reference selects magnetics and output capacitor. The primary system’s accepted design matches the tenth reconstructed magnetics candidate; none of the first eight is healthy. This is a current-code reconstruction, not a historical rank: the primary log ranks that candidate first after health filtering. Route selection and candidate truncation are thus distinct obstacles. These records do not isolate an option-cap effect.

Table S9: Verified task counts in the component ablation at $R = 0$. The main figure reports the group-weighted rates and paired differences.

Configuration	Single /72	Coupled /12	Natural /12
Full PowerFix	61	10	6
w/o Failure characterization	31	3	4
w/o Graph-guided diagnosis	36	1	0
w/o Constrained repair generation	44	3	2
w/o Isolated validation	59	10	5

Resampling implementation. The primary intervals use 100,000 bootstrap samples. Single and coupled tasks share a draw of twelve input–output specifications; natural tasks are resampled separately. The backbone intervals use 10,000 paired draws of the twelve recorded base-design groups. Each group carries six single, one coupled, and one natural task together across models. Both procedures recompute the three group rates in each draw, preserving the pairing for method differences.

Simulation counters. *Workflow simulations* are the calls logged by each repair workflow, used in the primary comparison and budget study. For PowerFix they include failure reproduction and candidate validation, but exclude external

scoring. *Logical operating-point evaluations* count requests at individual line/load/bus conditions, including cache hits. A sweep can contribute eight requests; settling extensions count separately. The selection and backbone studies use this counter. *Fresh executions* count PLECS runs that actually execute and exclude cache hits. These counters have different scopes and cannot be substituted for one another. Fresh counts and active repair time depend on the shared cache and execution order. The primary comparison lacks complete active-time records across all methods and tasks.

The token audit is in Appendix J; offline graph and authorization-map authoring is outside these per-task counters.

F Candidate and Replanning Budgets

Tables S10 and S11 give exact checkpoints behind the budget curves. A limits candidate validations within the initial plan; R permits additional planning rounds after validation feedback. A round may validate several candidates. The $A = 6$ and $R = 0$ rows reproduce the primary configuration’s group counts.

Table S10: Candidate limit under the initial plan ($R = 0$). Group columns give raw verified counts; VRSR is group-weighted.

A	Single /72	Coupled /12	Natural /12	VRSR (%)
1	61	8	4	61.6
2	64	9	4	65.7
3	66	10	5	72.2
4	67	10	6	75.5
6	68	11	6	78.7

Table S11: Replanning checkpoints. S/C/N are verified counts out of 72/12/12. Costs are group-weighted means per task, including failures; LLM calls exclude provider retries as separate calls.

R	S/C/N	VRSR (%)	Workflow simulations	Fresh executions	LLM calls
0	68/11/6	78.7	22.13	22.13	1.00
1	69/11/7	81.9	26.94	26.17	1.21
2	70/11/8	85.2	31.06	29.61	1.39
3	71/11/9	88.4	33.92	31.97	1.54
5	71/11/9	88.4	35.79	33.61	1.77

The grids are reported checkpoints. Omitting $R = 4$ does not mean the runner skips that round. These curves report point estimates; confidence intervals are not shown.

Group weighting gives one additional natural repair six times the contribution of one additional single-fault repair. The unchanged coupled count at later checkpoints does not establish an exhaustive search or identify why each remaining task stops. These budget results also do not provide an equal-resource comparison with the primary baselines.

G Repair-Dimension Selection

Policy definitions. Each policy selects two distinct permitted dimensions per planning round. The nine dimensions are turns ratio, magnetizing inductance, magnetic realization, MOSFET, rectifier, output capacitor, RCD clamp, input stage, and switching frequency. The fixed policy uses this priority: MOSFET, output capacitor, rectifier, magnetic realization, RCD clamp, turns ratio, magnetizing inductance, input stage, switching frequency. After a failure it selects an untried permitted pair. The random policy samples untried permitted pairs using the task-specific seed and round index. GPT-5 Mini reads the diagnosis, permitted dimensions, validation feedback, and previously failed pairs.

Task and seed counts. The 31 tasks admit multiple repair dimensions: seventeen single-fault, six coupled-fault, and eight natural failures. Five seeds give each policy 155 trajectories, with group denominators 85, 30, and 40. Table S12 retains both raw counts and VRSR because they weight coverage differently. At $R = 1$, for example, the fixed policy verifies more trajectories than random selection but has lower VRSR; more of its successes lie in the largest group.

Table S12: Verified trajectories, with group-weighted VRSR (%) in parentheses. Each policy has 155 trajectories.

Policy	$R = 0$	$R = 1$	$R = 2$	$R = 3$
Fixed	55 (27.4)	80 (39.4)	95 (54.7)	100 (60.2)
Random	50 (26.9)	78 (44.0)	93 (53.9)	118 (69.2)
GPT-5 Mini	91 (53.7)	125 (77.4)	133 (81.7)	136 (84.7)

Table S13: Group-weighted resource use per trajectory at $R = 3$, including failures. Simulation counters follow Appendix E.

Policy	Logical evaluations	Fresh executions	Active time (s)
Fixed	61.32	55.02	124.6
Random	61.12	55.09	124.6
GPT-5 Mini	40.15	37.43	162.6

The policy curves report point estimates; no significance tests are reported. Active time includes planning and depends on the execution conditions.

Separate full-system reference. Table S14 records full PowerFix on the same subset, with one trajectory per task. Group counts are 15/5/6 at $R = 0$ and 16/5/7 thereafter, out of 17/6/8. This reference allows the full admissible repair space, while Top-2 restricts selection to two dimensions. The one-versus five-seed protocols make the comparison descriptive; isolating the effect of that restriction requires matched seeds and candidate budgets.

Table S14: Full-system reference on 31 tasks, without seed replication. Costs are group-weighted means per task.

R	VRSR (%)	Tokens (k)	Logical evaluations	Fresh executions	Active time (s)
0	82.2	2.74	22.57	22.57	95.2
1	88.3	3.22	26.49	25.87	110.5
2	88.3	3.56	29.00	28.00	120.6
3	88.3	3.79	30.87	29.54	127.6

H Backbone Robustness

The first replicate contains 1,152 trajectories: 96 tasks for each of twelve models. An initial plan and up to five replanning rounds allow at most six candidate validations, one per round, with up to eight options per repair dimension. Search stops when a candidate is accepted. Table S15 reports initial success, final single-fault coverage, and recorded resource use under this protocol. VRSR@1 uses only the first candidate. The final GPT-5 Mini reference is 75.9%, as in Figure 7.

Attempt budgets and paired comparisons. The recorded checkpoints allow at most one, two, three, four, or six candidates. These are budget limits, not the number each task requires. After acceptance, later checkpoints retain the accepted outcome without running another repair. The table separates initial selection from the coverage obtained after replanning.

Qwen 3.7 Max and DeepSeek V4 Pro match GPT-5 Mini’s final binary outcome on every task in the first replicate. Their paired bootstrap differences are therefore zero in every re-sample. These zero-width intervals describe agreement on this task set; they do not remove uncertainty about other tasks or decoding runs. Grok 4.1 has the same group totals as GPT-5 Mini but differs on two tasks, giving a nonzero interval. The eleven alternative-model intervals shown in the main text are separate, unadjusted 95% intervals, not a simultaneous 95% guarantee for all comparisons.

Resource accounting. Simulation counters follow Appendix E. Fresh counts and active repair time depend on the shared cache and execution order; they describe these runs rather than intrinsic model speed.

Tokens include recorded input and output usage. API attempts include provider retries, whereas a workflow-level LLM request can contain several such attempts. GPT-5 Mini averages 2.40 API attempts and 2.34 workflow-level requests per task. Appendix J gives the prompt and token accounting. No complete official billing export is available, so the token column is not a comparison of monetary cost.

Replicate stability. The first replicate uses seed 20260721. A second replicate, with seed 20260722, covers a fixed subset of 24 tasks: twelve single-fault, six coupled, and six natural tasks. This adds 288 trajectories, for 1,440 in total. Every model has identical final binary outcomes across the two replicates on this subset. First-acceptance indices also agree on all 24 tasks for ten models, and on 23 of 24 for GPT-5 Mini and Grok 4.1; joint nonacceptance counts as agreement. This checks repeatability on the selected subset, not

Table S15: Outcomes and costs in the backbone study (first replicate). VRSR@1 is group-weighted success after the first candidate; Single@6 gives raw verified counts at the six-candidate limit. All models verify 10/12 coupled and 7/12 natural tasks at that limit. Resource columns are group-weighted means per task at the maximum budget, including failed tasks.

Model	VRSR@1 (%)	Single @6	Logical OP evaluations	Fresh executions	Tokens (k)	API attempts	Active time (s)
ERNIE 4.5 Turbo	68.5	63/72	18.05	3.92	18.55	2.38	104.3
GPT-5.4 Nano	68.5	63/72	18.31	2.58	15.15	2.35	79.1
Qwen 3.7 Max	68.5	62/72	18.78	3.06	19.16	2.36	125.3
DeepSeek V4 Pro	69.0	62/72	18.14	3.84	16.43	2.44	115.7
Grok 4.1	68.5	62/72	18.18	3.81	15.09	2.55	157.8
GPT-5 Mini	68.1	62/72	18.70	4.21	17.46	2.40	148.5
Qwen 3.5 Flash	68.5	61/72	18.72	3.75	25.82	2.52	170.3
Doubao Seed 1.6	68.5	61/72	18.31	3.69	19.79	2.36	140.9
GLM-5.2	69.0	61/72	18.76	2.99	20.13	2.57	185.0
Claude Sonnet 5	65.7	61/72	18.25	3.58	23.88	2.41	144.7
Claude Opus 5	63.0	61/72	19.03	6.00	30.25	2.92	273.8
Gemini 3.6 Flash	68.5	59/72	18.76	3.86	20.66	2.40	108.6

on all 96 tasks. It does not establish stability under provider updates or identify the model snapshots missing from historical records.

I Model-Access Provenance

The audit distinguishes requested aliases from gateway-returned identifiers. For the primary coupled and natural batch (24 tasks across four methods), runtime fingerprints confirm a shared endpoint, <https://api.vveai.com/v1>, and requested alias `gpt-5-mini`. Returned identifiers include both `gpt-5-mini` and `gpt-5-mini-2025-08-07`; these strings alone do not establish different upstream snapshots.

The backbone study uses the same gateway with model-specific aliases; its GPT-5 Mini condition requests `gpt-5-mini-2025-08-07`. Historical probes, condition fingerprints, and request metadata support that routing provenance. Endpoint records are missing for the single-fault batch and the earlier ablation and budget runs, which predate the logging configuration. Those comparisons are matched at the documented model-family level. Establishing a shared upstream snapshot across batches would require provider records we do not hold.

J Token Usage and Request Audit

The audit covers 480 task results: four primary methods on 96 tasks and the 96-task GPT-5 Mini backbone condition, with 2,084 workflow-level call records. All result files match their frozen manifest or table hashes. This confirms record identity, not upstream model identity (Appendix I). Appendix F gives the budget checkpoints.

Counting usage. Table S16 includes unsuccessful tasks. Tokens per request divide group-weighted per-task tokens by requests, before rounding. Usage may include retries within a request; missing usage is not estimated. These online costs exclude graph and map authoring and deterministic enumeration. They are not total ownership costs or an equal-budget comparison.

Table S16: Recorded token usage. Req./API gives workflow requests and API attempts per task; API attempts include retries. Input and output are tokens per request; total is thousands of tokens per task. PowerFix conditions follow Table 1.

Condition	Req./API	Input	Output	Total
PowerFix (primary)	1.00/1.06	1,895.96	835.97	2.73
PowerFix (backbone)	2.34/2.40	6,215.58	1,238.18	17.46
Open-loop	4.00/4.44	6,535.81	2,738.94	37.10
Self-Refine	5.23/5.29	5,134.95	1,257.65	33.44
Tool Agent	11.08/11.32	6,865.63	613.14	82.86

What the prompts contain. The primary planner receives check identifiers, numerical diagnosis, evidence, and registered repair routes. It orders routes without further LLM calls at $R = 0$.

Open-loop roles repeatedly receive requirements, the failed design, and failure information, with catalogs or earlier proposals where relevant. Self-Refine analysis reads the full design, catalogs, and feedback; edit parsing receives editable paths. The Tool Agent reads the design, editable fields, tools, remaining budget, and last eight tool-history entries.

The backbone planner reads the full design, failures, diagnosis, and four dimensions: MOSFET, rectifier, output capacitor, and magnetics. It selects one or two without expanding candidate lists in the prompt. Later rounds add validation feedback and the last three attempts. This differs from primary route ordering and nine-dimension selection. Initial backbone planning averages 6,048.97 input and 1,023.13 output tokens; later calls average 6,339.68 and 1,398.34. Later calls occur only on tasks that have not succeeded, so this difference cannot isolate feedback overhead.

K Accepted Patch Profile

We profile all 85 accepted task submissions in the primary $R = 0$ configuration: 68 single-fault, 11 coupled-fault, and

6 natural-failure tasks. Each joint repair counts as one submission; only the final selected route is included. Rejected candidates are not part of the field-count distribution. Percentages use these 85 submissions as the denominator, without equal-group weighting.

Counting fields. For each submission, we count distinct field paths explicitly present in `primary_edits`, taking their union across constituent patches. A write is counted even when its recorded old and new values agree. The measure therefore describes the recorded write footprint, rather than the number of values that differ between the initial and final design. Rollback entries, read-only attributes, and downstream recomputed fields absent from the explicit edits are excluded. Table S17 gives every nonzero frequency; the JSON also retains the zero-frequency bins. Quantiles use linear interpolation at index $(n - 1)q$, with $n = 85$ and $q = 0.9$ for P90. A threshold share is the sum of counts at or below that threshold divided by 85.

Table S17: Distribution of explicit field writes in accepted $R = 0$ submissions. Each field path counts once per submission, including writes that retain its current value.

Explicit fields	1	2	3	25	26
Submissions	28	13	3	34	7

Write composition. The *swap* class contains only writes to the MOSFET, rectifier, output-capacitor, or magnetic-core part-number fields: 28 submissions (32.9%). The *parameter* class contains no such reference writes: zero submissions. The *mixed* class contains both reference and other field writes: 57 submissions (67.1%). These labels describe the explicit patch contents, not independently verified physical replacements. Composition is assigned after taking the union of writes in a joint submission, rather than separately for each atomic patch.

Operation coverage. The experimental catalog registers four atomic operators (Table S18). These are distinct from the seven topology-level repair-operation nodes in Power-DiagGraph (Appendix D). An operator is counted once per submission in which it appears. Ten submissions combine two operators, so the frequencies sum to 95 rather than 85.

Table S18: Atomic operators in accepted $R = 0$ submissions. Joint repairs contribute to more than one row.

Experimental operator	Submissions
<code>swap_mosfet</code>	20
<code>swap_rectifier</code>	18
<code>swap_output_capacitor</code>	16
<code>redesign_magnetics</code>	41

Validation and rollback. No explicit field path falls outside the experimental authorization map, and all 85 accepted submissions (100%) contain rollback edits. This checks the

presence of rollback records, not a separate rollback execution. Before the selected repair was accepted, 83 submissions had no validation-rejected candidate and two had one. This count includes earlier routes but excludes compilation failures, preselection failures, and checks of alternative routes after acceptance. Repeated historical trace events are excluded using the executed-candidate counter and current-run timestamps. It is consistent with $R = 0$, which permits candidate validation within the initial plan but no feedback-driven replanning. This profile covers $R = 0$ only.